

SAILABLENDRA AKASH BONAGIRI

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San Francisco Bay Area, CA

SUMMARY

PhD Student in Computer Science with 4+ years building ML systems across LLMs, AI Agents, and NLP. Expert in **LLM post-training, LLM safety & Privacy, AI Agent Evals, RAG, PEFT/LoRA fine-tuning**, Published researcher with peer-reviewed papers (incl. **Best Paper Award**, ICML, ACM CHI, AAAI) and industry experience deploying ML at scale.

EDUCATION

University of California, Davis

Sep 2022 – Jun 2027

PhD in Computer Science (*Research focus: Applied ML, AI Agent Evaluations*)

Davis, California

- Coursework: Machine Learning, Advanced Deep Learning, NLP, Web & Data Analytics, Distributed Databases

Birla Institute of Technology and Science, Pilani (BITS PILANI)

Aug 2017 – May 2021

Bachelor of Engineering in Electronics and Communication

Hyderabad, India

EXPERIENCE

Lead Machine Learning Researcher – PRISM AI Group

Sep 2024 – May 2026

Tech: Python, LangChain, LangGraph, RAG, MCP, LLM-as-a-Judge, AWS SageMaker

California, USA

- Architected **end-to-end Automated Agent Eval Pipelines** developing a unique **LLM-as-a-Judge** architecture to ensure reliability, improve performance, and optimize costs, cutting evaluation turnaround by **85%**
- Integrated **red-teaming** and **A/B testing**, uncovering **critical failure modes** across agentic workflows
- Launched **ResearchQuest AI**, a **RAG**-powered agent with **CoT** reasoning, reducing literature review time by **70%**
- Engineered a Multimodal Medical AI Agent for explainable diagnosis generation enhancing diagnostic accuracy by **25%**
- Trained deep learning **anomaly detection** models on large-scale sensor data improving detection accuracy by **20%**

Applied Machine Learning Researcher – C2 AI Lab

Sep 2022 – Sep 2024

Tech: Pytorch, RAG, LangChain, Transformers, Multimodal LLMs, GCP, PEFT, LoRA

California, USA

- Applied **PEFT** & **LoRA** fine-tuning on LLMs for harm detection, achieving **30%** accuracy gain over **SOTA**
- Fine-tuned **Multimodal LLMs** using **SFT** and **ICL** for **few-shot** classification across **3 video platform** datasets
- Developed **NLP4GOV(RAG + CoT)** library reducing search latency by **60%** while boosting precision by **25%**

Machine Learning Engineer – PepsiCo

Jun 2021 – Sep 2022

Tech: TensorFlow, FastAPI, Streamlit, Azure, Docker, Kubernetes, Hadoop, MLOps

Hyderabad, India

- Led **Precision-Agro**, a **multimodal SaaS ML** app deployed at scale, driving a **35%** increase in crop yield
- Reduced Time to Conversion by **40%** via a field image diagnosis app serving **500+ agricultural workers**
- Designed **recommendation system** validated with **A/B testing**, improving Gross Margin by **15%**

Machine Learning Intern – Innoviti Payment Solutions

Jul 2020 – Dec 2020

Tech: Python, Predictive Models, Recommendation Systems, Time Series, EDA

Bengaluru, India

- Developed **time series forecasting** models to predict merchant trends, improving loyalty algorithm efficiency by **20%**
- Crafted ML-driven **personalization models** for fintech merchant preferences, boosting **customer retention** by **35%**

SELECTED PUBLICATIONS

- “**STABLEVAL**: Disagreement-Aware and Stable Evaluation of AI Systems.” In **ICML**
 - “**Safer Social Platforms**: Scalable Few-Shot Harm Moderation Using LLMs.” In **AAAI ICWSM**
 - “**NLP4Gov**: A Comprehensive Search & Retrieval Library for Computational Policy Analysis.” In **CHI**
 - “**Aletheia**: Fake News Detection System for Hindi.” In **ACM IKDD CODS COMAD (Best Paper Award)**
- Full list on [Google Scholar](#), Reviewed Top-tier AI, ML conference papers at NeurIPS, ICML, AAAI, CHI, ICA etc.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, R

Agent Frameworks: LangChain, LangGraph, MCP, AI Agent Builder, AutoEval, TerminalBench

MLOps & Infra: MLOps, MLflow, Model Monitoring, AWS SageMaker, AWS, GCP, Azure, Docker, Kubernetes, Elasticsearch, FastAPI, Flask, PostgreSQL, RESTful API, React, GitHub Actions CI/CD

ML & AI: TensorFlow, Pytorch, Scikit-learn, XGBoost, HuggingFace, Transformers, LLMs, NLP, Deep Learning, RAG, LLM-as-a-Judge, PEFT, LoRA/QLoRA, SFT, RLHF, Prompt Engineering, Chain-of-Thought, Multimodal Models, Computer Vision, Red-teaming, OpenAI Evals, Statistical Modeling, Hypothesis Testing, Feature Engineering, PySpark, OpenCV, NLTK, numpy, pandas, matplotlib